226183: sandstone, Nicolay 1

(Nambeet Formation, Canning Basin)

Location and sampling

Joanna Spring (SF 51-3), McTavish (3556)
MGA Zone 51, 526953E 7825419N

Sampled on 18 February 2019

This sample was collected from the 3236.0 – 3236.5 m depth interval of drillcore Nicolay 1, from a petroleum well drilled in 2012 by New Standard Onshore Pty Ltd (New Standard Onshore, 2013). The drillhole is located on a north–south cleared line, about 52 km south of Battlement Rocks, 30 km southeast of Turkey Place Hill, and 18.8 km west-northwest of McTavish Claypan.

Geological context

The unit sampled is the Early Ordovician (485–470 Ma; Cohen et al. 2020) Nambeet Formation of the Canning Basin (Haines, 2004). The unit is about 337 m thick in Nicolay 1 and consists mainly of shale with interbedded limestone and sandstone (New Standard Onshore, 2013). The Nambeet Formation is the lowermost Canning Basin unit in Nicolay 1 and is conformably overlain by the Willara Formation (Haines, 2004; Zhan, 2019). Three additional samples of Nambeet Formation sandstone from Nicolay 1 yielded maximum depositional ages of 828–562 Ma (GSWA 226184, Wingate et al., 2021a; GSWA 226185, Wingate et al., 2021b; GSWA 226186, Wingate et al., 2021c). A Nambeet Formation sandstone sample from Samphire Marsh 1, drilled about 246 km to the west-northwest, yielded detrital zircon ages of 2838–506 Ma, and a conservative maximum depositional age of 519 ± 3 Ma (GSWA 135962, Wingate and Haines, 2009). Another Nambeet Formation sample from Barnicarndy 1, drilled about 185 km to the southwest, yielded detrital zircon ages of 3274–590 Ma, and a conservative maximum depositional age of 1007 ± 18 Ma (GSWA 237998, Wingate et al., 2021d).

Petrographic description

The sample is a very fine-grained sandstone interbedded with wavy layers of clay-rich siltstone (Fig. 1). The sandstone component consists of about 65% quartz, 20–25% feldspar, <10% matrix, 2% pyrite, 1–2% detrital mica, 1–2% fossil fragments and trace tourmaline, chalcopyrite and zircon. The sandstone contains small wisps of brown silt or clay and the laminated clay-rich siltstone layer contains discontinuous sandstone lenses within brown siltstone. Quartz is mainly subangular, clear, unstrained, and up to 0.15 mm in diameter. Feldspar is subangular, up to 0.15 mm in size, partially clay-altered, and consists of both plagioclase and microperthite. Detrital micas include subhedral laths to 0.2 mm long of both muscovite and chlorite, the latter possibly after biotite. Straight to curved narrow rods, mainly in the sandstone, consist of either carbonate or brown collophane, and are interpreted as fossil fragments. Rounded grains of yellow–green tourmaline and zircon, both up to 0.1 mm in diameter, also occur in the sandstone. Pyrite occurs throughout the sample as single cubic grains up to 0.1 mm across and in aggregates up to 0.2 mm in size. The siltstone component is dominated by clay and iron oxide minerals, with about 5% quartz and minor feldspar, detrital mica, and finer-grained pyrite. Bedding is moderately sharp but wavy in form. The sandstone component is well sorted although the grains have generally not been rounded. This, and the abundance of clay-rich matrix suggests that the sediment lacks textural maturity or represents a textural inversion. The presence of minor fossil debris indicates a marine environment. There is no evidence of deformation or metamorphism.

Zircon morphology

Zircons isolated from this sample are mostly colourless to pale brown, anhedral to euhedral, and mainly strongly rounded. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 4:1. Most crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11 June 2020, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 8900s quadrupole ICP-MS. Samples were ablated for 22 s at a 5 Hz repetition rate, using a 24 µm beam diameter and laser energy of 2.0 J/cm^2. Zircon standard 91500 was used as the primary reference material, and GJ1, OGC1 and Plešovice were included as secondary standards. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Ages from analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

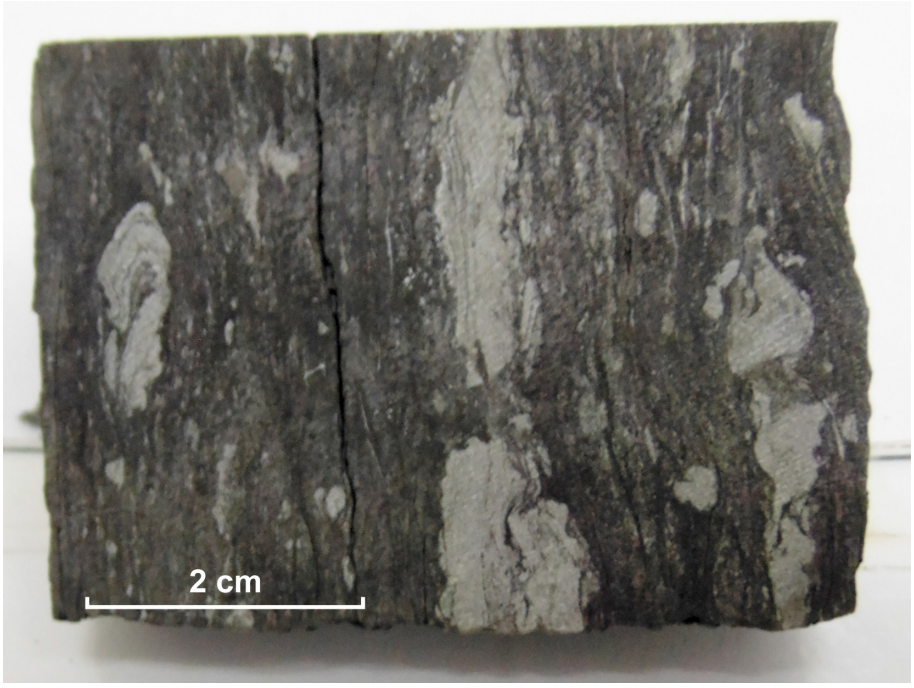

Figure 1. Drillcore image of sample 226183: sandstone, Nicolay 1

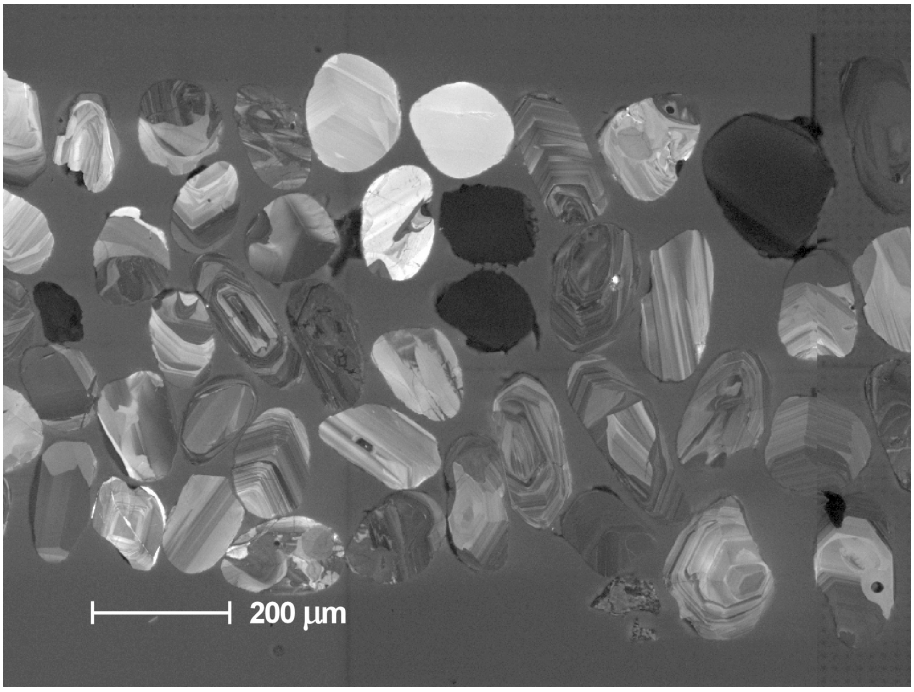

Figure 2. Cathodoluminescence image of representative zircons from sample 226183: sandstone, Nicolay 1

Table 1. LA-ICP-MS analytical results for zircons from sample 226183: sandstone, Nicolay 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
Y	67	67	80	77	0.95	13.141	0.121	0.05590	0.00225	473	4	448	89	-5.4
S	47	47	125	58	0.46	12.469	0.093	0.05950	0.00165	497	4	585	60	15.1
S	13	13	283	160	0.56	12.115	0.081	0.05740	0.00125	511	3	507	48	-0.9
S	33	33	119	76	0.63	12.019	0.101	0.05790	0.00165	515	4	526	62	2.1
S	11	11	253	199	0.79	11.719	0.069	0.05740	0.00125	528	3	507	48	-4.1
S	19	19	350	226	0.65	11.616	0.067	0.05810	0.00120	532	3	534	45	0.2
S	63	63	128	100	0.78	11.261	0.095	0.06670	0.00205	548	4	828	64	33.8
S	64	64	1481	515	0.35	11.136	0.062	0.05611	0.00075	554	3	457	30	-21.4
S	18	18	113	132	1.16	11.038	0.091	0.05830	0.00180	559	4	541	68	-3.3
S	42	42	111	1	0.01	11.013	0.091	0.05870	0.00170	560	4	556	63	-0.8
S	39	39	202	152	0.75	10.830	0.065	0.05910	0.00120	569	3	571	44	0.2
S	23	23	13	49	3.61	10.787	0.209	0.05700	0.00550	571	11	492	213	-16.3
S	49	49	42	75	1.78	10.471	0.132	0.05800	0.00255	588	7	530	96	-11.0
S	54	54	304	125	0.41	10.428	0.065	0.06070	0.00110	590	4	629	39	6.1
S	60	60	568	410	0.72	10.195	0.062	0.05870	0.00095	603	4	556	35	-8.5
S	43	43	1097	204	0.19	10.028	0.050	0.05950	0.00080	613	3	585	29	-4.7
S	38	38	695	222	0.32	10.017	0.048	0.06020	0.00095	613	3	611	34	-0.4
S	14	14	238	69	0.29	9.542	0.059	0.06190	0.00120	642	4	671	41	4.2
S	44	44	365	141	0.39	9.422	0.053	0.06130	0.00110	650	3	650	39	-0.1
S	40	40	1287	449	0.35	8.781	0.046	0.06392	0.00080	695	3	739	26	5.9
S	52	52	562	217	0.39	7.353	0.041	0.07660	0.00115	822	4	1111	30	26.0
S	32	32	1287	97	0.08	7.062	0.040	0.07038	0.00085	854	5	939	25	9.1
S	28	28	74	95	1.29	6.631	0.048	0.07500	0.00175	905	6	1069	47	15.3
S	9	9	364	170	0.47	6.464	0.036	0.07210	0.00110	927	5	989	31	6.2
S	6	6	195	86	0.44	6.341	0.042	0.06850	0.00130	944	6	884	39	-6.8
S	36	36	482	118	0.25	6.109	0.062	0.07140	0.00110	977	9	969	31	-0.9
S	50	50	175	54	0.31	6.090	0.043	0.07300	0.00130	980	6	1014	36	3.3
S	51	51	39	27	0.68	5.974	0.050	0.07120	0.00230	998	8	963	66	-3.6
S	62	62	624	138	0.22	5.804	0.030	0.07330	0.00100	1025	5	1022	28	-0.2
S	55	55	339	393	1.16	5.627	0.035	0.07420	0.00105	1054	6	1047	29	-0.7
S	53	53	274	115	0.42	5.397	0.032	0.07430	0.00110	1096	6	1050	30	-4.4
S	41	41	544	369	0.68	5.593	0.031	0.07430	0.00100	1060	5	1050	27	-1.0
S	10	10	143	117	0.82	5.519	0.038	0.07450	0.00135	1074	7	1055	36	-1.8
S	37	37	331	118	0.35	5.362	0.027	0.07490	0.00115	1102	5	1066	31	-3.4

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$\frac{^{232}Th}{^{238}U}$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
S	29	29	370	222	0.60	5.373	0.030	0.07610	0.00115	1100	6	1098	30	-0.2
S	27	27	208	187	0.90	5.040	0.028	0.07800	0.00130	1167	6	1147	33	-1.7
S	68	68	187	353	1.89	5.336	0.061	0.07800	0.00135	1107	12	1147	34	3.5
S	34	34	131	108	0.82	4.995	0.032	0.07830	0.00145	1176	7	1154	37	-1.9
S	8	8	43	32	0.74	5.214	0.048	0.07840	0.00235	1131	9	1157	59	2.2
S	7	7	272	161	0.59	5.359	0.027	0.07850	0.00115	1103	5	1160	29	4.9
S	16	16	756	329	0.44	5.020	0.023	0.07890	0.00100	1171	5	1170	25	-0.1
S	48	48	228	193	0.85	4.900	0.029	0.07900	0.00120	1197	6	1172	30	-2.1
S	31	31	84	95	1.13	5.086	0.038	0.07990	0.00185	1157	8	1195	46	3.1
S	35	35	341	338	0.99	4.697	0.025	0.08160	0.00115	1244	6	1236	28	-0.7
S	2	2	442	343	0.78	4.468	0.021	0.08300	0.00110	1302	6	1269	26	-2.6
S	5	5	603	598	0.99	4.456	0.023	0.08340	0.00105	1305	6	1279	25	-2.1
S	1	1	737	154	0.21	4.623	0.022	0.08529	0.00105	1262	6	1322	24	4.5
S	20	20	850	86	0.10	3.817	0.018	0.09247	0.00110	1500	6	1477	23	-1.6
S	58	58	103	74	0.72	3.849	0.027	0.09320	0.00160	1489	9	1492	32	0.2
S	3	3	44	45	1.01	3.982	0.033	0.09430	0.00215	1444	11	1514	43	4.6
S	56	56	85	139	1.63	3.167	0.020	0.10780	0.00170	1769	10	1763	29	-0.4
S	15	15	113	49	0.44	3.122	0.017	0.10960	0.00170	1791	9	1793	28	0.1
S	30	30	302	54	0.18	3.026	0.015	0.11140	0.00140	1841	8	1822	23	-1.0
S	24	24	579	289	0.50	3.070	0.014	0.11160	0.00135	1818	7	1826	22	0.4
S	12	12	779	131	0.17	2.598	0.012	0.13290	0.00150	2099	8	2137	20	1.8
S	59	59	144	82	0.57	2.230	0.013	0.15910	0.00205	2388	12	2446	22	2.4
S	61	61	359	220	0.61	1.940	0.009	0.18500	0.00220	2680	10	2698	20	0.7
D	66	66	55	50	0.91	5.698	0.052	0.07670	0.00200	1042	9	1113	52	6.4
D	17	17	226	145	0.64	4.771	0.024	0.07730	0.00115	1227	6	1129	30	-8.7
D	21	21	234	136	0.58	5.144	0.030	0.08040	0.00125	1145	6	1207	31	5.1
D	4	4	233	142	0.61	4.929	0.024	0.08270	0.00130	1191	5	1262	31	5.6
D	26	26	1410	409	0.29	4.411	0.029	0.08967	0.00105	1317	8	1419	22	7.1
D	22	22	237	179	0.75	3.656	0.021	0.10200	0.00140	1559	8	1661	25	6.2
D	46	46	928	291	0.31	3.337	0.020	0.11300	0.00130	1690	9	1848	21	8.6
D	65	65	356	179	0.50	3.531	0.018	0.11650	0.00145	1607	7	1903	22	15.5
D	57	57	307	157	0.51	3.109	0.020	0.11690	0.00150	1798	10	1909	23	5.9
D	45	45	278	135	0.49	2.747	0.038	0.15500	0.00190	2001	24	2402	21	16.7
D	25	25	496	213	0.43	1.911	0.009	0.16990	0.00195	2713	11	2557	19	-6.1

Results

Sixty-eight analyses were obtained from 68 zircons. Results are listed in Table 1, and shown in a concordia diagrams (Fig. 3) and probability density plots (Figs 4, 5).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Eleven analyses >1000 Ma are >5% discordant. The dates obtained from these 11 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 57 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 473 ± 4 Ma (1σ).

Group S comprises 56 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or 207-corrected $^{238}U/^{206}Pb^*$ dates of 2698–496 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 473 ± 4 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean 207-corrected $^{238}U/^{206}Pb^*$ date of 533 ± 16 Ma (MSWD = 3.4) for the youngest coherent group of three analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 1817, 1484, 1266, 1161, 1053, 998, 980, 590, 568, and 557 Ma, based on contributions from approximately 4, 3, 5, 8, 8, 7, 3, 3, 4, and 4 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

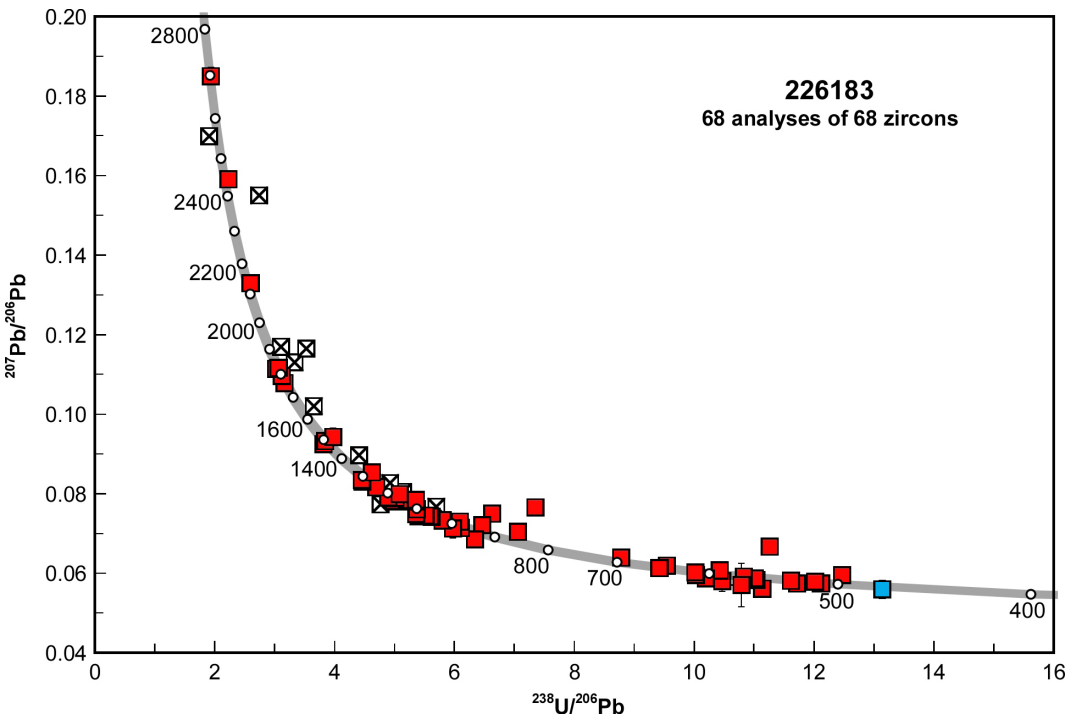

Figure 3. U–Pb analytical data, not corrected for common Pb, for zircons from sample 226183: sandstone, Nicolay 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma and discordance >5%)

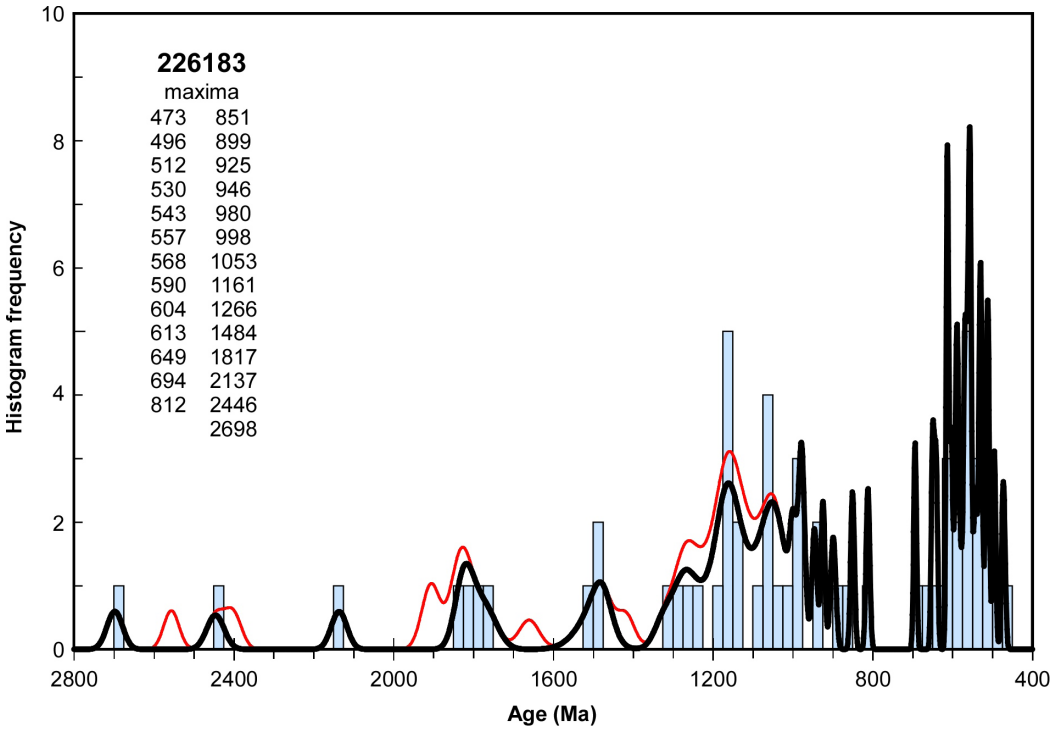

Figure 4. Probability density diagram and histogram for sample 226183: sandstone, Nicolay 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) include only data in Groups Y and S (57 analyses of 57 zircons). Thin curve includes all data (68 analyses of 68 zircons)

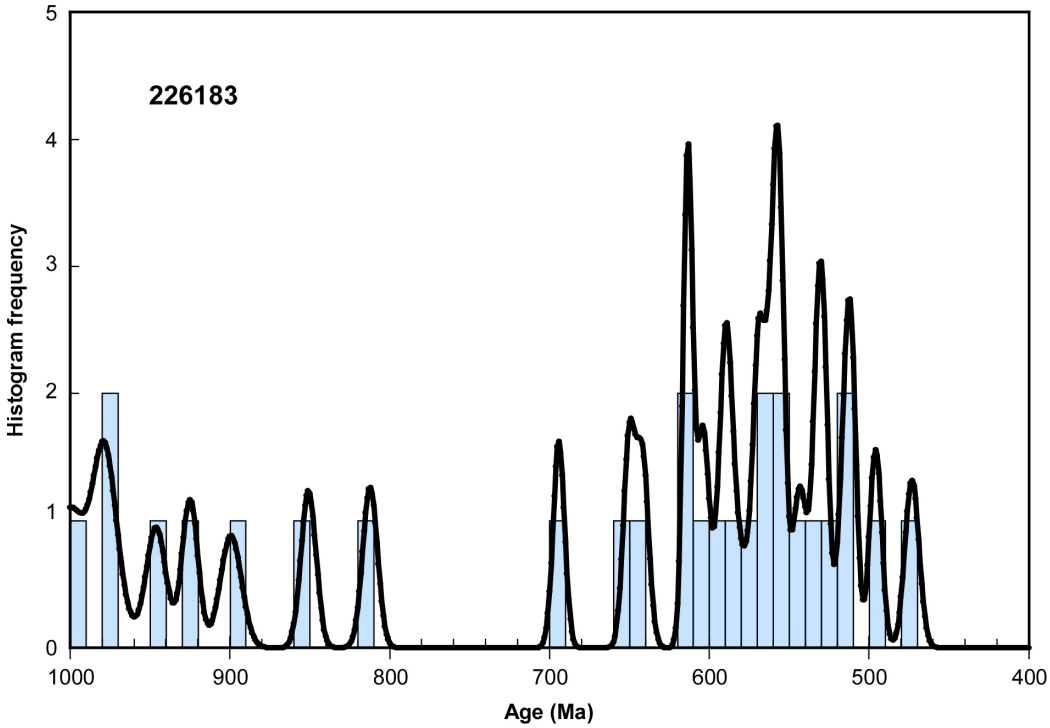

Figure 5. Expanded view of probability density diagram and histogram for sample 226183: sandstone, Nicolay 1. Bin width is 10 Ma

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Recommended reference for this publication

Wingate, MTD, Lu, Y and Dent, LM 2021, 226183: sandstone, Nicolay 1; Geochronology Record 1760: Geological Survey of Western Australia, 7p.

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